T2 TECHNOLOGIES

Feeder Balancing Module (FBM) Tool — Tender Enquiry 3239CXMWP

Author work package — Thabiso

This brief consolidates every section of the tender submission draft assigned to Thabiso, with the current draft content reproduced in full so each section can be developed independently before being merged back into the master document.

Assignment Summary

Section	Title	Ownership
5	Solution Architecture and Design Principles	Owned by Thabiso
6	Delivery Methodology and Implementation Plan	Owned by Thabiso
7	Programme Governance and Quality Management	Owned by Thabiso
8	Security and Compliance	Owned by Thabiso
9	Architecture Deep-Dive	Owned by Thabiso
17	Commercial Proposal Framework / Pricing Schedule	Shared — Tumelo / Thabiso

5. Solution Architecture and Design Principles

Owned by Thabiso

The target architecture will be confirmed during discovery, but T2 Technologies will apply the following principles unless the Client's standards require otherwise.

Principle	Application
Modularity	Separate business capabilities into clear modules and interfaces so that components can evolve independently.
API First	Expose controlled interfaces for system interaction instead of direct database coupling wherever practicable.
Security by Design	Apply identity, authorisation, encryption, secrets management, auditability and secure coding from the start.
Traceability	Use correlation identifiers and structured logs to trace transactions across service boundaries.
Resilience	Apply timeouts, retries where safe, circuit-breaking or isolation patterns, failure handling and recovery controls.
Idempotency	Protect critical write operations from accidental duplication when business or integration flows can be retried.
Observability	Capture metrics, logs and service health to support proactive operational management.
Data Governance	Define ownership, quality controls, retention, access and lineage for critical data.
Automation	Automate builds, tests, deployments and repeatable infrastructure tasks to reduce operational error.
Portability	Avoid unnecessary technology lock-in and preserve the ability to evolve the deployment model.

5.1 Illustrative Logical Architecture

- Experience Layer: web portals, mobile applications, administration interfaces and assisted-service channels.
- API / Integration Layer: authenticated APIs, routing, orchestration, validation, throttling, transaction controls and external connectors.
- Business Services Layer: domain-specific services, workflows and business rules.
- Data Layer: operational databases, integration stores, reporting data and controlled analytics pipelines.
- Platform Layer: runtime environments, containers/virtual infrastructure, network controls, secrets, backup and recovery.
- Operations Layer: CI/CD, monitoring, logging, alerting, vulnerability management, service management and audit reporting.

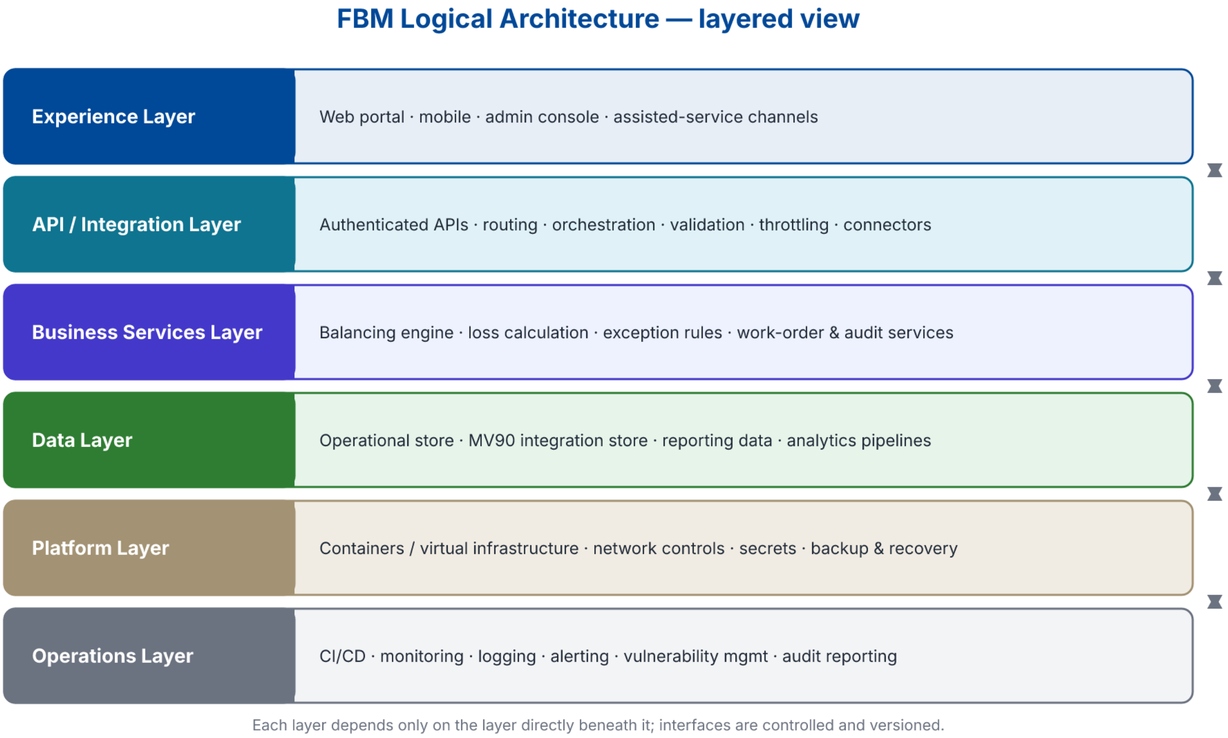


Figure 5.1 — Logical architecture

5.2 Reference Deployment and Data Flow

The module is designed for a resilient, cloud-ready deployment. Traffic is terminated at a managed edge, routed to stateless application services behind a private network boundary, and backed by managed data and integration stores. The data-flow view shows how metered readings move through import, validation and balancing into the decisions and actions the tool surfaces.

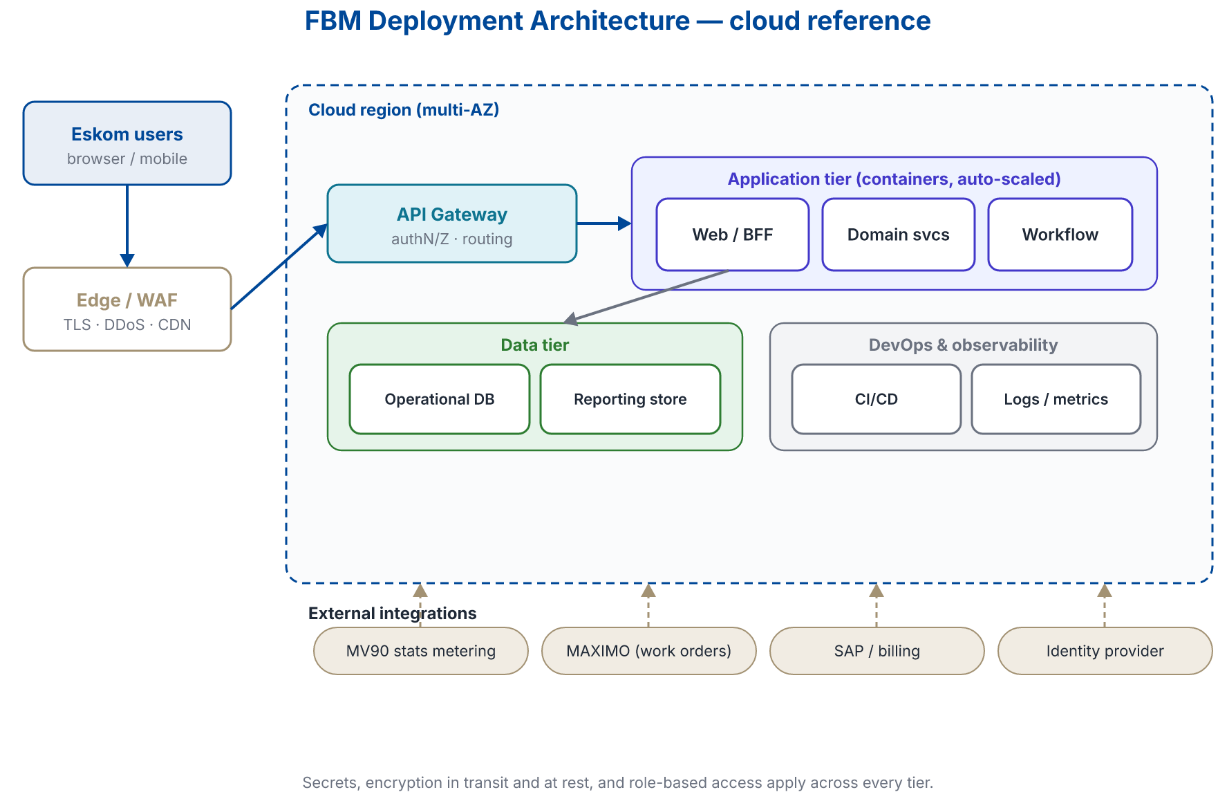


Figure 5.2 — Deployment architecture

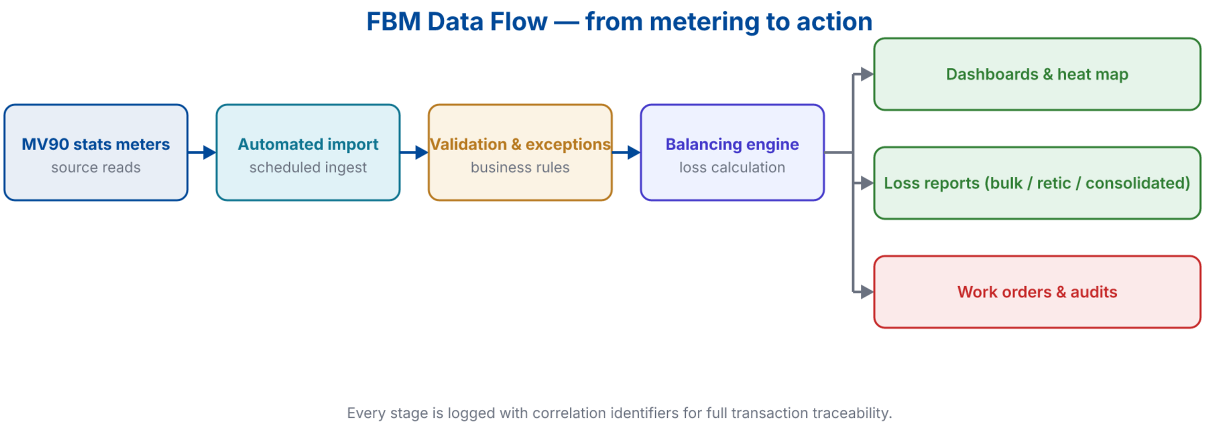


Figure 5.3 — Data flow

5.3 Application Configuration Workflows

The following workflows explain how the tool is operated day to day — behaviour that cannot be conveyed by a static screen alone. Each workflow diagram is paired with a capture of the corresponding screen in the live FBM application.

Configuring a network establishes the eight-level hierarchy (Operating Unit down to Meter); network mapping then lets an engineer re-parent equipment and group meters, transformers or feeders into clusters by drag-and-drop, with energy roll-ups recalculated automatically.

Workflow — Configuring the network hierarchy

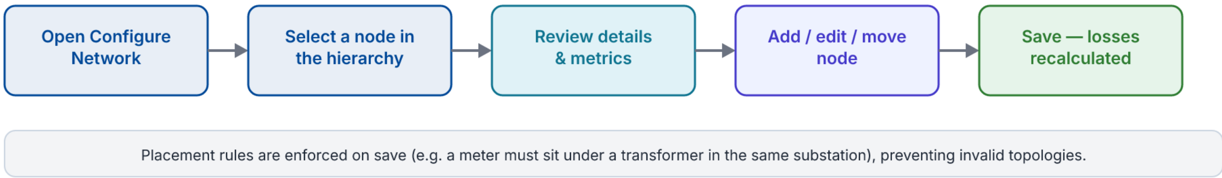


Figure 5.4 — Network configuration workflow

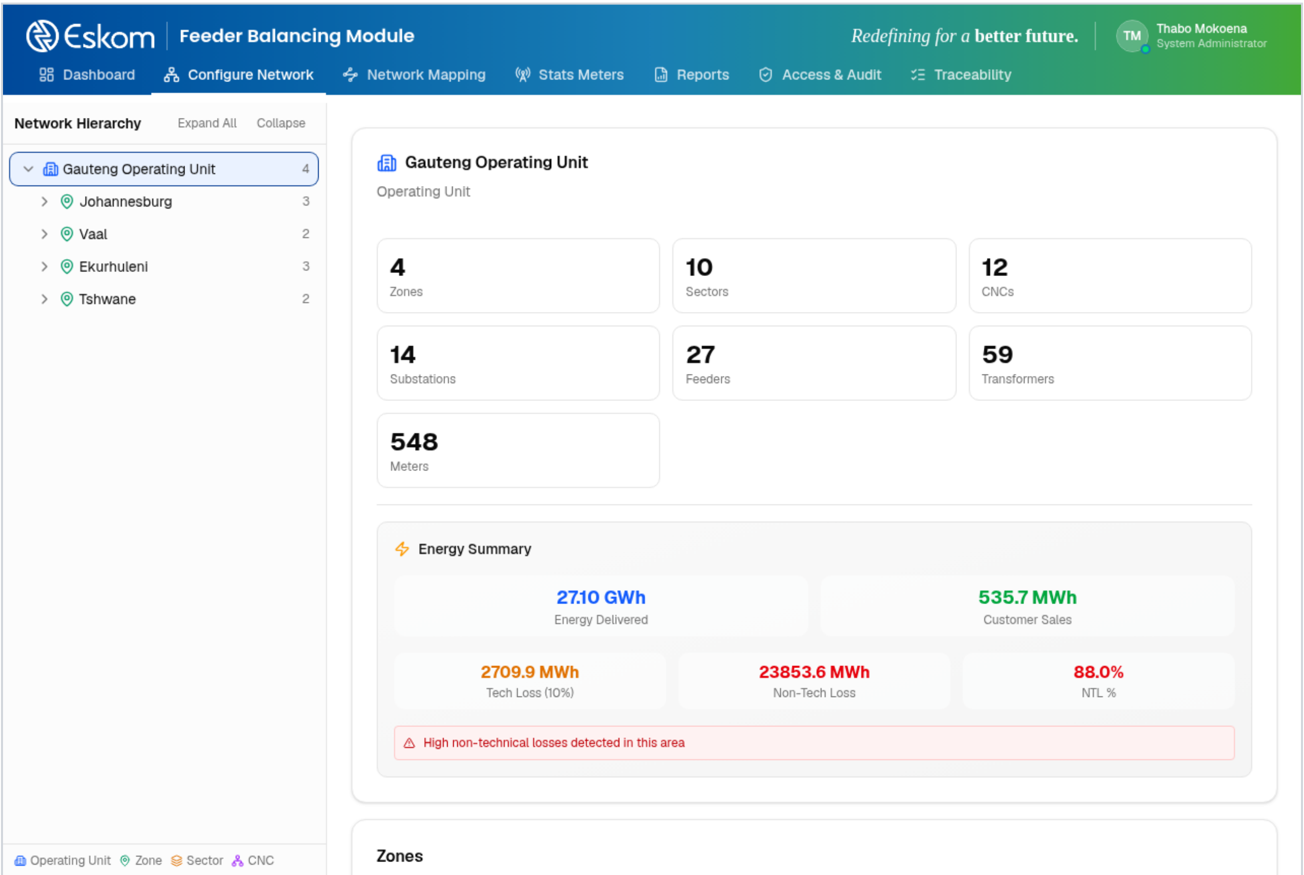
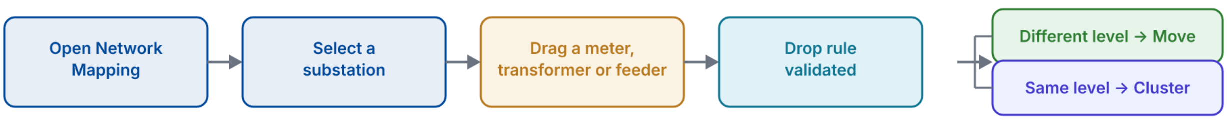


Figure 5.7 — Configure Network (live tool): the eight-level hierarchy with automatic energy roll-up and loss KPIs computed at every level.

Workflow — Network mapping (drag to reorganise)



Drops are validated live; only same-substation, rule-compliant moves are accepted.

Figure 5.5 — Network mapping workflow

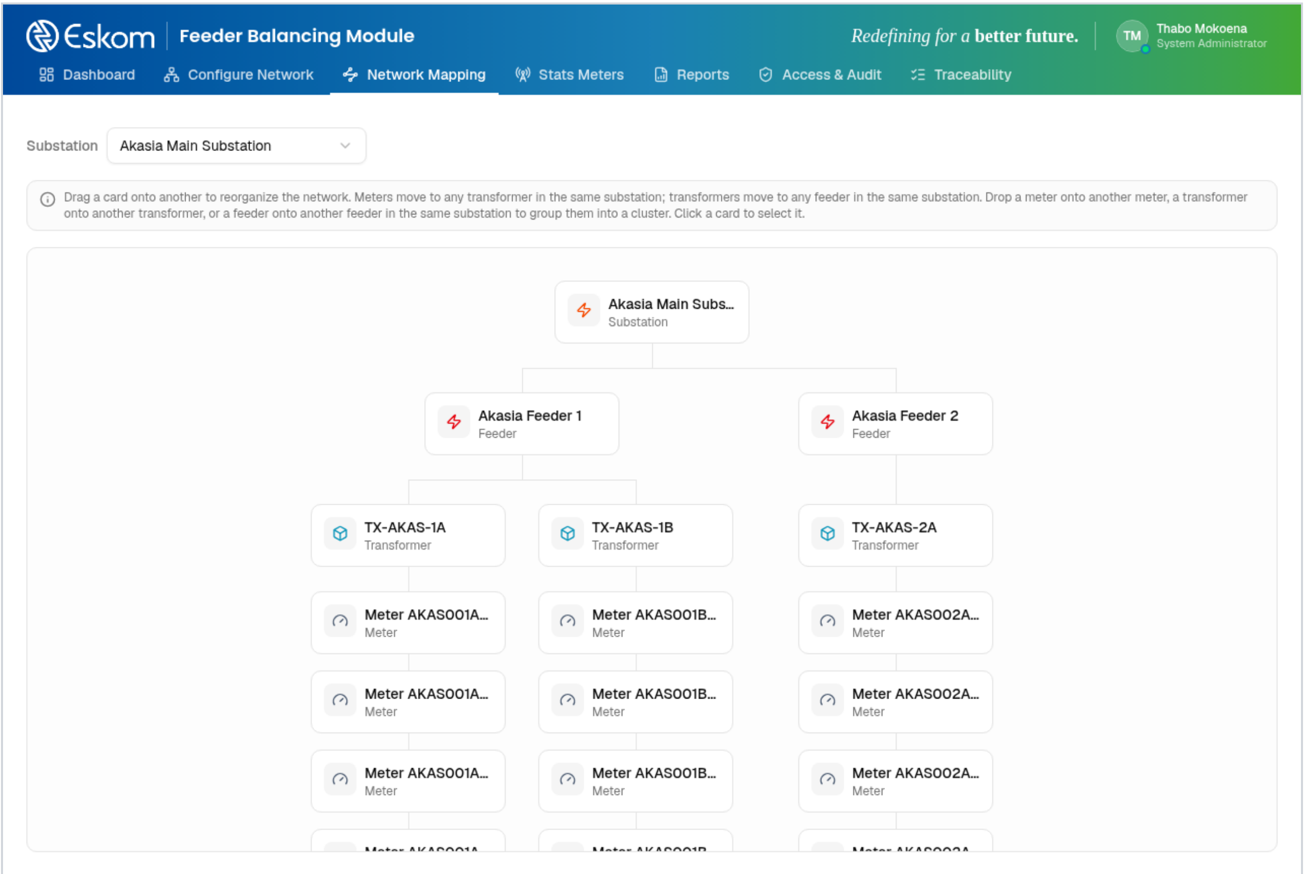


Figure 5.8 — Network Mapping (live tool): drag-and-drop organisation chart for a selected substation.

Workflow — Creating a meter cluster

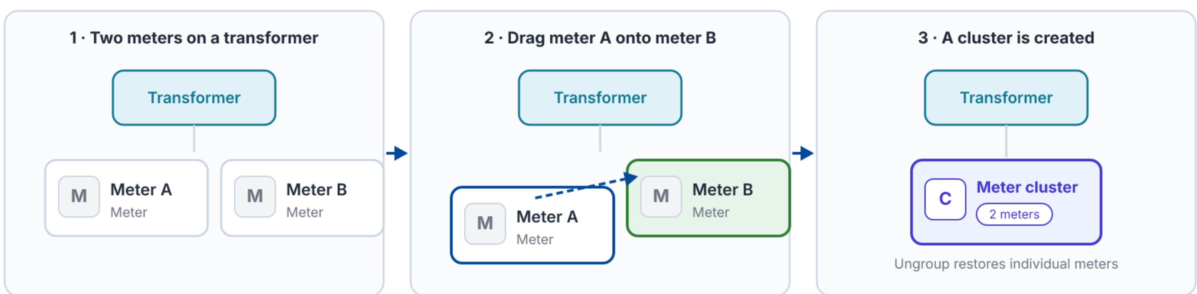


Figure 5.6 — Creating a meter cluster

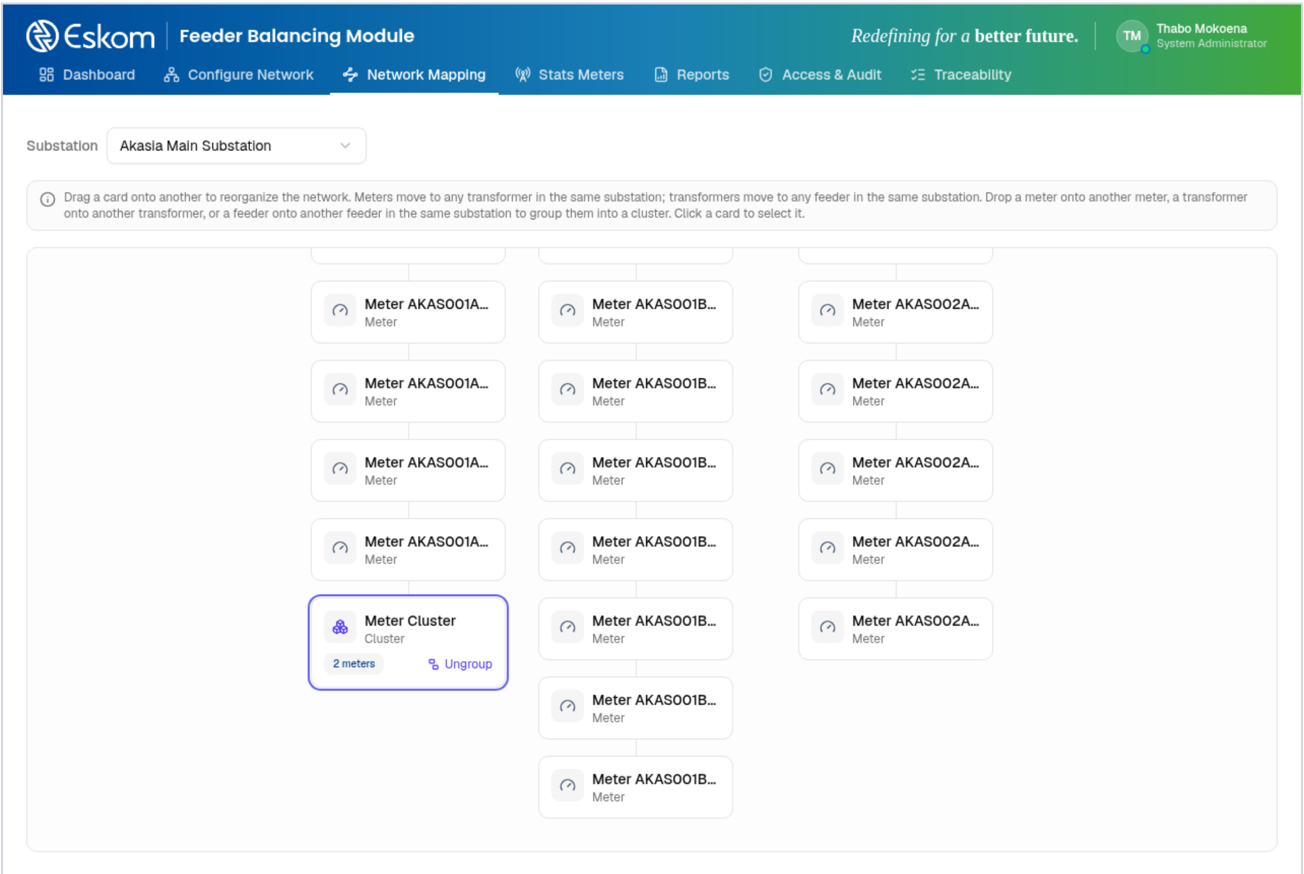


Figure 5.9 — Network Mapping (live tool): two meters grouped into a meter cluster, with one-click Ungroup to reverse the grouping.

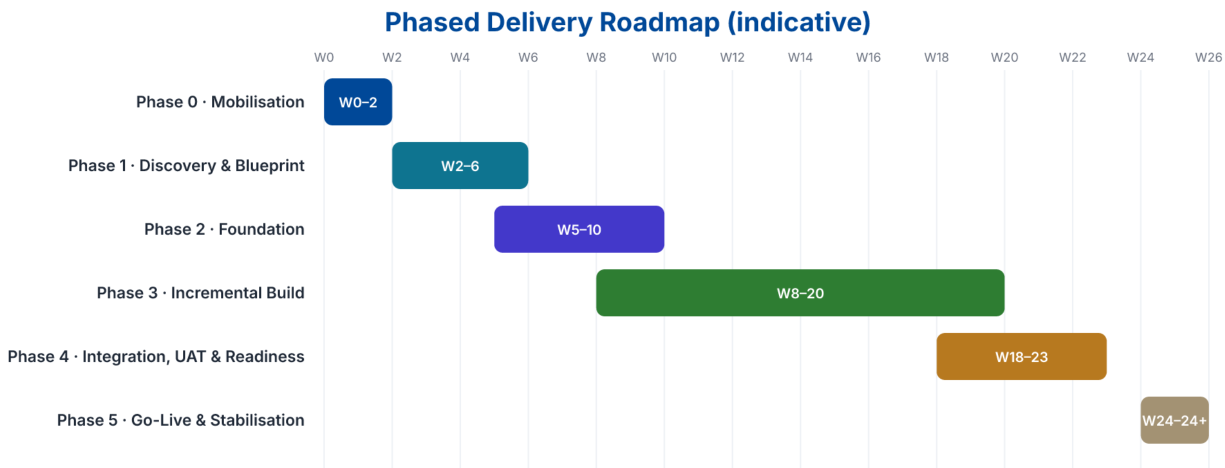
6. Delivery Methodology and Implementation Plan

Owned by Thabiso

T2 Technologies proposes a phased agile delivery model governed by formal stage gates. This balances iterative product development with the traceability and accountability expected in enterprise and public-sector programmes.

Phase	Indicative Timing	Key Activities	Exit Outcome
Phase 0 - Mobilisation	Weeks 1-2	Contract mobilisation, governance, access, stakeholder map, delivery plan, risks, environments and document baseline.	Approved mobilisation pack and detailed project plan.
Phase 1 - Discovery & Blueprint	Weeks 2-6	Current-state assessment, requirements, process mapping, architecture, security assessment and prioritisation.	Requirements baseline, target architecture, backlog and migration approach.
Phase 2 - Foundation	Weeks 5-10	Core platform, CI/CD, security baseline, integration foundation, environments and observability.	Operational technical foundation ready for solution increments.
Phase 3 - Incremental Build	Weeks 8-20	Applications, APIs, integrations, workflows, data pipelines and automated tests delivered in sprints.	Demonstrable functional releases with traceability.
Phase 4 - Integration, UAT & Readiness	Weeks 18-23	End-to-end testing, performance/ security testing, user acceptance, training, migration rehearsal and operational readiness.	UAT sign-off and go-live readiness approval.
Phase 5 - Go-Live & Stabilisation	Weeks 24+	Production deployment, hypercare, defect resolution, monitoring, support transition and post-implementation review.	Stable production service and transition to BAU support.

The above timeline is illustrative. Th
volume, data migration complexity,



Illustrative only — final schedule reconciles to issued tender milestones, scope volume and dependencies.

Figure 6.1 — Delivery roadmap

Sprint Delivery Cycle

Figure 6.2 — Sprint cycle

6.1 Sprint Delivery

- Prioritised product backlog linked to approved requirements.
- Regular planning, refinement, demonstrations and retrospectives.
- Definition of Done covering functional behaviour, testing, security, documentation and deployability.
- Formal change control for scope or contractual impacts outside the approved backlog baseline.
- Release notes and auditable evidence for each production release.

7. Programme Governance and Quality Management

Owned by Thabiso

Governance will provide clear authority, reporting, escalation and evidence-based decision making. T2 Technologies recommends a joint structure that includes executive oversight, programme management and technical working groups.

Forum	Purpose	Indicative Cadence
Steering Committee	Executive decisions, scope/budget oversight, strategic risks, escalations and milestone approval.	Monthly or as prescribed
Programme Review	Progress, dependencies, RAID log, commercial status, upcoming milestones and actions.	Weekly
Architecture / Security Review	Design decisions, integration standards, security controls, technical debt and exceptions.	Weekly / per major design
Sprint Ceremonies	Planning, stand-ups, demonstrations, retrospectives and backlog refinement.	Per sprint
Service Review	SLA performance, incidents, problems, capacity, releases and continuous improvement.	Monthly post go-live

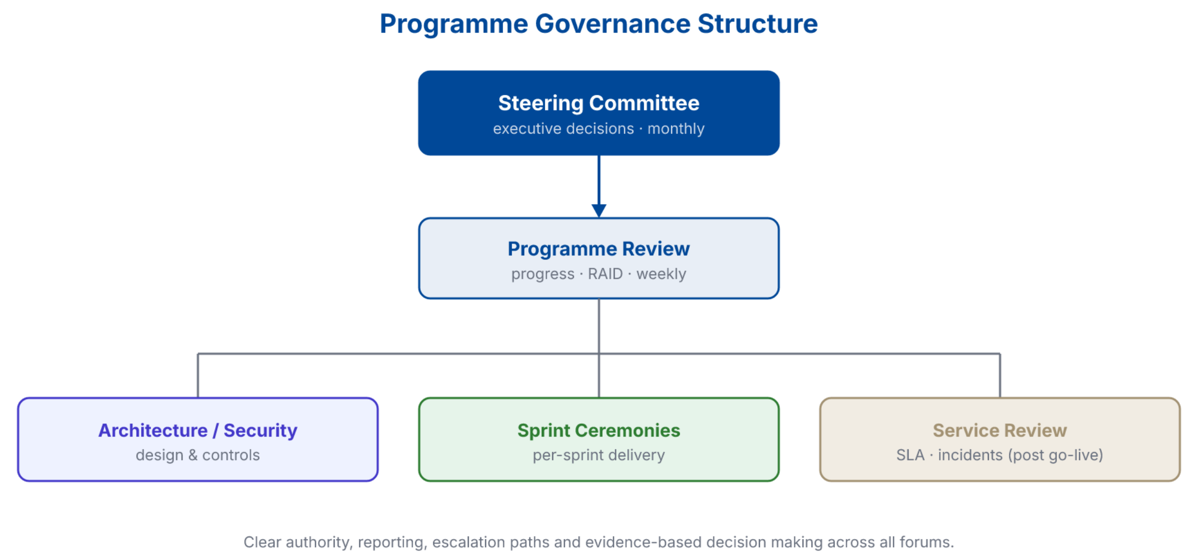


Figure 7.1 — Governance structure

7.1 Quality Controls

- Requirements-to-test traceability.
- Peer review and controlled source-code management.
- Automated unit/integration testing wherever practical.
- Security checks integrated into the development lifecycle.
- Controlled environments and release approvals.
- Document versioning and approval records.
- Defect prioritisation and transparent closure evidence.
- Post-release verification and service health monitoring.

8. Security and Compliance

Owned by Thabiso

Security is treated as a design property of the FBM solution rather than an afterthought. The architecture applies defence-in-depth: independent, mutually reinforcing controls at the perimeter, network, identity, application and data layers, so that the failure of any single control is contained by the layers around it. This section responds to the non-functional security requirements of the Statement of Work (SoW 5.1) and is designed to align with Eskom's information-security standards, which will be confirmed during discovery.

The controls below describe the intended security posture for the production service. They are stated as design commitments subject to detailed design, Client security standards and the outcome of a joint threat and risk assessment; specific certification scopes are noted as assumptions where relevant.

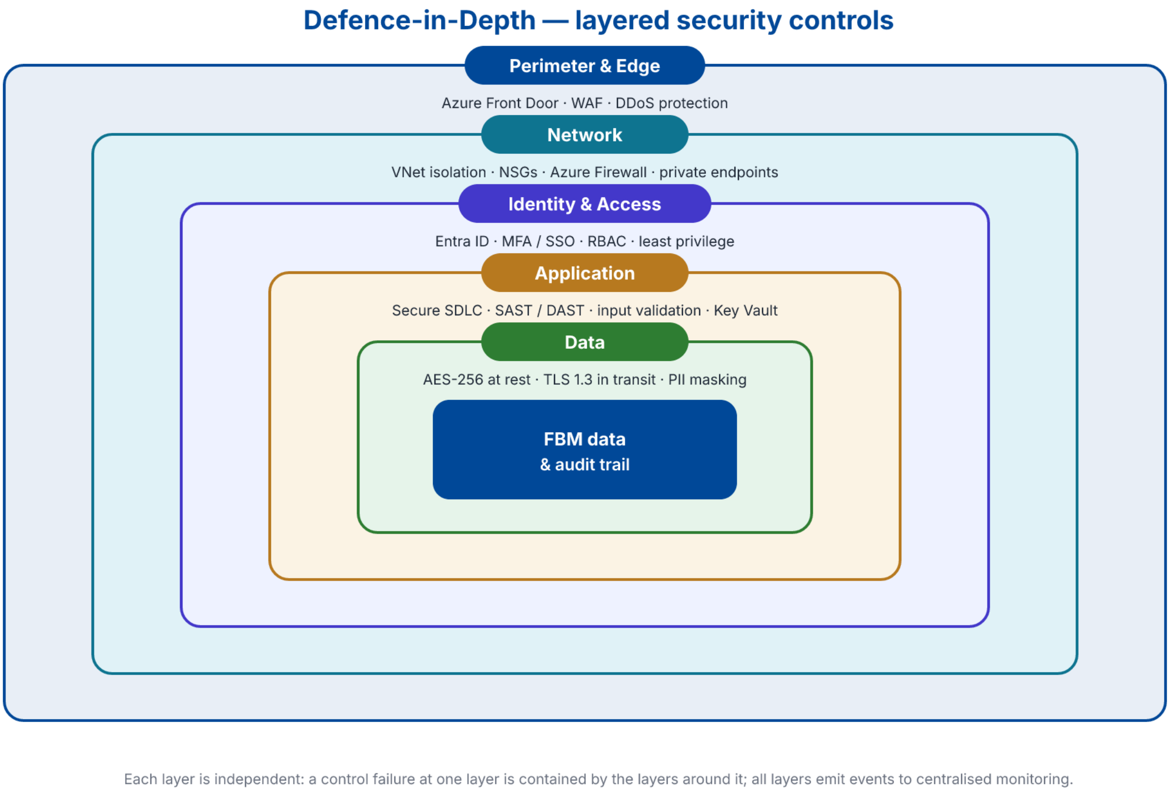


Figure 8.1 — Defence-in-depth security layers

8.1 Security Principles

- Secure by design and by default — security requirements are captured in the backlog and verified as part of the Definition of Done.
- Least privilege and need-to-know — access is granted against defined roles and reviewed periodically.
- Defence-in-depth — no single control is relied upon; controls are layered and independent.
- Zero-trust orientation — every request is authenticated, authorised and logged regardless of network location.
- Encryption everywhere — data is protected in transit and at rest across all tiers.
- Segregation of duties and environments — development, test and production are isolated with controlled promotion.

8.2 Identity and Access Management

Authentication federates to Eskom's identity provider so that FBM never holds primary credentials. Every request passes identity, edge and authorisation controls before reaching application services, and each decision — grant or deny — is written to the immutable audit trail already demonstrated in the prototype.

- Single sign-on via Microsoft Entra ID (Azure AD) with multi-factor authentication enforced for all users.
- Role-based access control mapped to the FBM capability model demonstrated in the application (e.g. view, edit network, manage meters, resolve exceptions, initiate audits, export, view audit log).
- Privileged Identity Management with just-in-time elevation and approval for administrative roles.
- Joiner-Mover-Leaver process integration so access follows the user lifecycle.
- Periodic access recertification and automatic session expiry.

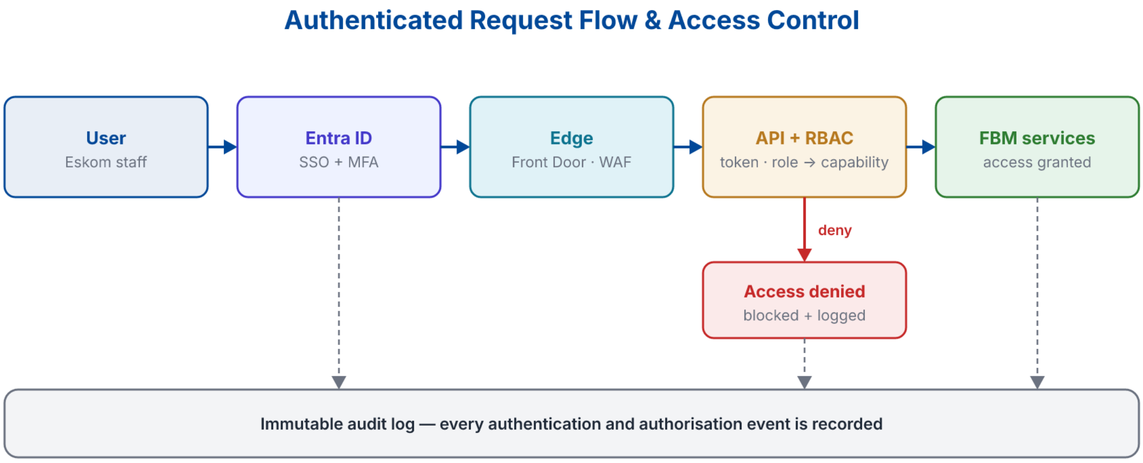


Figure 8.2 — Authenticated request & access-control flow

8.3 Data Protection and Privacy

- Encryption at rest using AES-256 (Transparent Data Encryption for Azure SQL; server-side encryption for storage).
- Encryption in transit using TLS 1.3 for all external and service-to-service traffic.
- Data residency within the Republic of South Africa (Azure South Africa North), addressing data-sovereignty requirements.
- Personally identifiable information masked or de-identified in non-production environments.
- Defined data-classification, retention and secure-disposal policies.
- POPIA-aligned processing, including lawful basis, data-subject rights handling and records of processing.

8.4 Application and Platform Security

- Secure SDLC with threat modelling, secure-coding standards and mandatory peer review.
- Static and dynamic application security testing (SAST/DAST) and software-composition analysis in the CI/CD pipeline.
- Container image scanning and hardened, patched base images.
- Secrets and keys held in Azure Key Vault — never in source control or configuration files.
- Web Application Firewall, DDoS protection and rate limiting at the edge.

- Independent penetration testing before every major production release and periodically thereafter.

8.5 Monitoring, Audit and Incident Response

- Centralised logging and SIEM via Azure Monitor, Log Analytics and Microsoft Sentinel.
- Immutable, tamper-evident audit trail of security-relevant events (demonstrated in the prototype's Access & Audit module).
- Real-time alerting on anomalous authentication, authorisation and data-access events.
- Documented incident-response runbooks with defined severities and escalation paths.
- POPIA breach-notification handling within the regulated timeframe, with e-discovery support.
- Regular control testing, log review and continuous-improvement feedback into the backlog.

8.6 Compliance and Standards Alignment

The solution is designed to support the following frameworks. The matrix shows which control domains each framework informs; certification scope and supporting evidence are to be confirmed with T2 Technologies and the Client.

Framework	Relevance to FBM
POPIA (South Africa)	Primary data-protection law governing processing of personal information.
ISO/IEC 27001	Information Security Management System baseline for controls and governance.
ISO/IEC 27017 & 27018	Cloud-specific security controls and protection of PII in the cloud.
SOC 2 Type II	Independent attestation of security, availability and confidentiality controls.
GDPR	Applied as good practice for privacy-by-design and data-subject rights.
NIST Cybersecurity Framework	Structures the overall posture: Identify, Protect, Detect, Respond, Recover.
Cybercrimes Act & King IV	South African statutory and governance context for security and oversight.

Where T2 Technologies does not currently hold a specific certification, the commitment is to o standard and to pursue certification within a timeframe agreed with the Client. All items are su discovery.

Compliance Framework Mapping

	POPIA	ISO/IEC 27001	SOC 2 Type II	GDPR	ISO 27017/18	NIST CSF
Data protection & privacy	✓	✓	✓	✓	✓	✓
Identity & access management	✓	✓	✓	✓	✓	✓
Encryption (rest & transit)	✓	✓	✓	✓	✓	✓
Logging, audit & monitoring	✓	✓	✓	✓	✓	✓
Secure development (SAST/DAST)	—	✓	✓	✓	—	✓
Resilience, backup & DR	—	✓	✓	—	✓	✓
Incident & breach response	✓	✓	✓	✓	—	✓

Mapping indicates design intent; certification scope and evidence to be confirmed with T2 Technologies and the Client during discovery.

Figure 8.3 — Compliance framework mapping

9. Architecture Deep-Dive

Owned by Thabiso

This section expands the solution architecture in Section 5 into an implementable, cloud-native target for deployment into Eskom's Microsoft Azure tenant, with all data resident in South Africa. The design favours composable, containerised microservices, an API-first integration model and elastic scaling, so the platform can grow from the demonstrated prototype to a national roll-out without re-architecture.

The specific Azure services named below are a reference design. Final service selection, sizing, region pairing and landing-zone configuration will be confirmed with Eskom's cloud and security teams during discovery.

Architecture is presented as a reference design aligned to the SoW's cloud, security and scalability requirements. Target tenant configuration, service selection and non-functional targets (availability, RPO/RTO, performance) must be confirmed with Eskom during discovery.

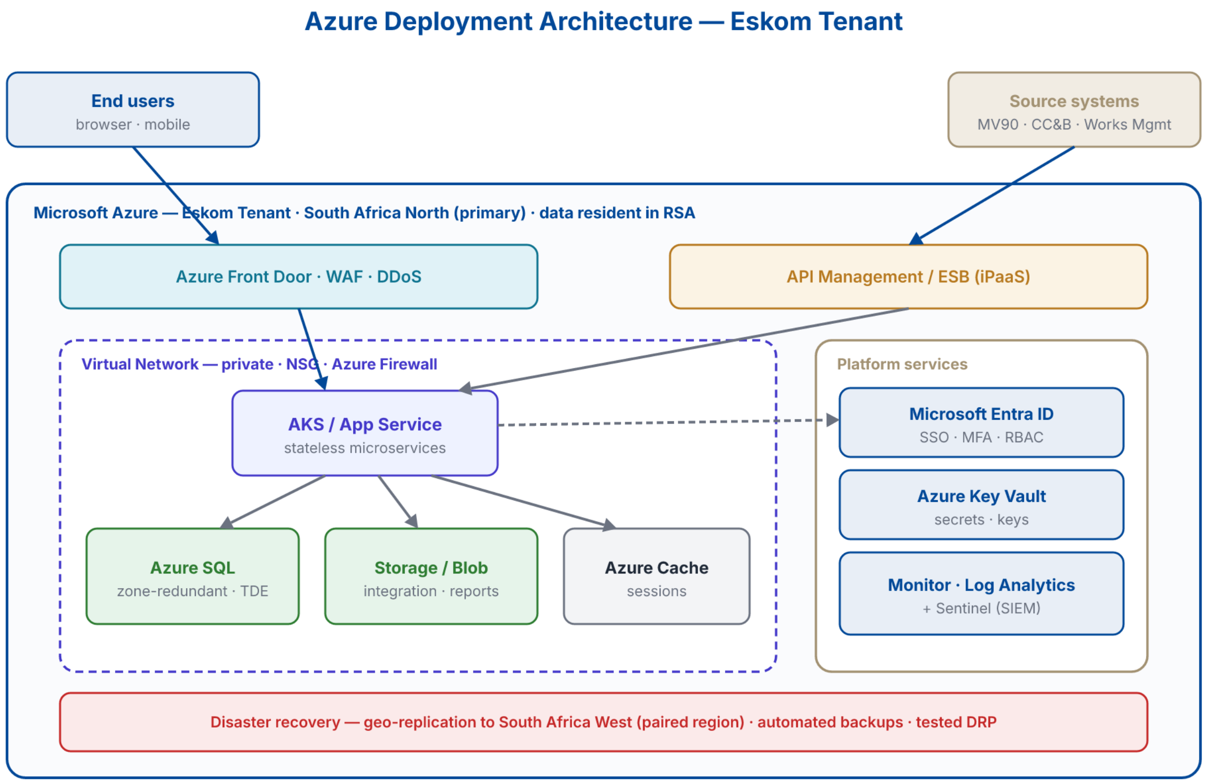


Figure 9.1 — Azure deployment architecture

9.1 Azure Landing Zone and Hosting

- Deployment into an Eskom-governed Azure landing zone with policy, RBAC and cost guardrails.
- Primary region Azure South Africa North (Johannesburg) for data sovereignty.
- Resource segregation by environment and workload using subscriptions and resource groups.
- Infrastructure as Code (Bicep or Terraform) for repeatable, auditable provisioning.
- Private networking by default — virtual networks, network security groups, Azure Firewall and private endpoints.

9.2 Application Architecture

- Composable microservices packaged as containers on AKS or Azure App Service.
- Stateless services that scale horizontally behind Azure Front Door and API Management.
- API-first design with versioned, documented contracts between services and channels.
- Event-driven ingestion for MV90 and other feeds, decoupled via queues for resilience.
- Autoscaling driven by demand, with graceful degradation under load.

9.3 Data Architecture

- Azure SQL (Business Critical, zone-redundant) as the operational store with Transparent Data Encryption.
- Separation of operational data from reporting/analytics data to protect transactional performance.
- Azure Storage/Blob for the integration store, imported files and generated report extracts.
- Azure Cache for sessions, rate-limiting state and hot lookups.
- Defined retention, archiving and point-in-time restore aligned to Client policy.

9.4 Integration Architecture

FBM integrates with upstream systems of record through a managed integration layer using a canonical data model, so that source-system change is absorbed by adapters rather than propagated into the core.

- API Management / Enterprise Service Bus (iPaaS) mediates all inbound and outbound integration.
- Source adapters for MV90 stats meters, CC&B billing, Works Management (e.g. MAXIMO) and CNL/GIS.
- Batch and near-real-time patterns with idempotent processing and dead-letter queues.
- Outbound publication to BI/corporate reporting and the Losses Reduction Potential tool.
- Correlation identifiers propagated across every hop for end-to-end traceability.

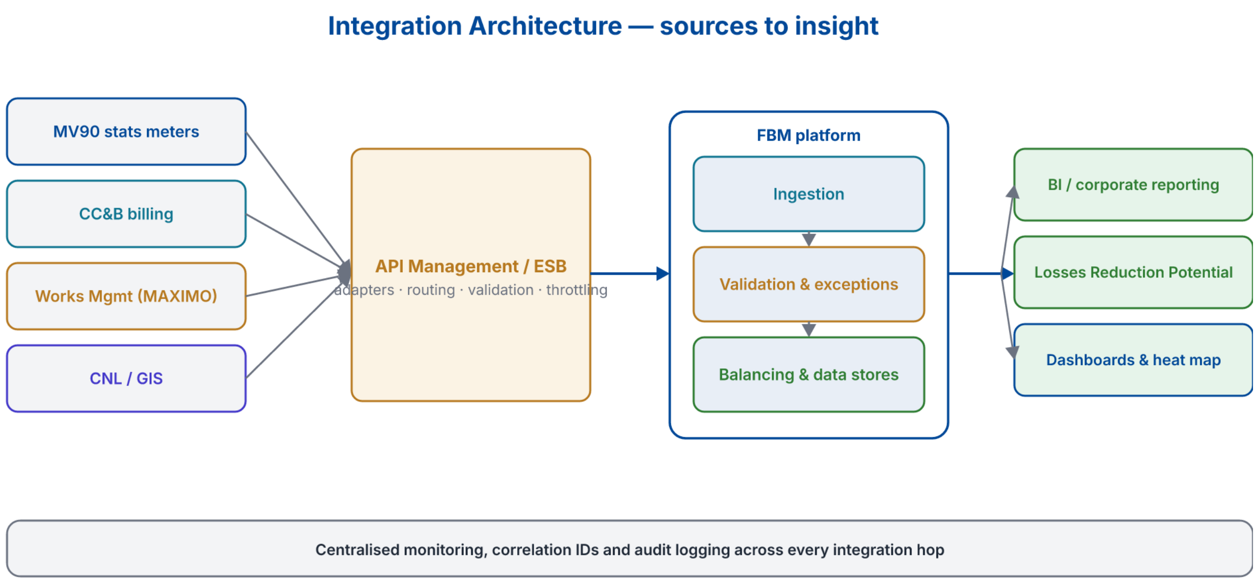


Figure 9.2 — Integration architecture

9.5 Scalability, Availability and Disaster Recovery

- Horizontal autoscaling of stateless services and zone-redundant data services for high availability.

- Target availability, RPO and RTO to be agreed and underpinned by the SLA.
- Geo-replication to the paired region (Azure South Africa West) for disaster recovery.
- Automated, encrypted backups with regular restore testing.
- A documented, regularly tested Disaster Recovery Plan (DRP).

9.6 Environments and DevOps

Identical, isolated environments are promoted through an automated pipeline with quality and security gates, giving predictable, auditable releases.

- Separate DEV, TEST/UAT, staging and PROD environments provisioned from the same IaC.
- CI/CD via Azure DevOps or GitHub Actions with build, automated test, SAST/DAST and release gates.
- Manual approval gate before production promotion, with full release evidence.
- Blue-green or canary release strategy to minimise deployment risk.
- Observability (metrics, logs, traces) wired in from day one.

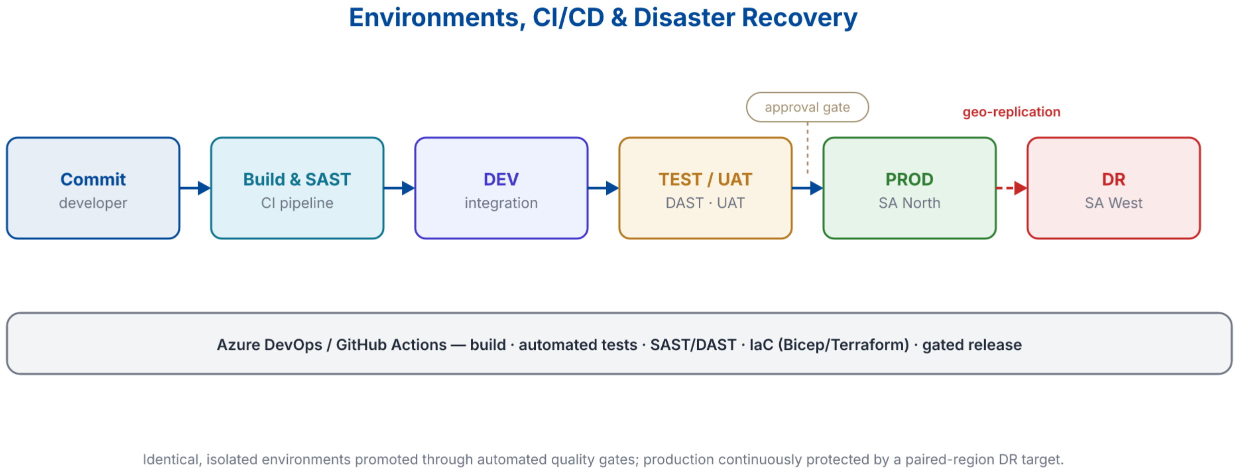


Figure 9.3 — Environments, CI/CD & DR

17. Commercial Proposal Framework / Pricing Schedule

Shared — Tumelo / Thabiso

This section presents a fully costed commercial estimate aligned to the structure of the tender's Pricing Schedule (Annexure L): a software/subscription licence line, itemised implementation work-packages, a professional-services rate card, support & maintenance, a five-year total cost of ownership and a milestone-based payment plan. All figures are in South African Rand and exclude VAT unless stated otherwise.

The estimate assumes an 18-month implementation of the business requirements followed by managed support & maintenance to the five-year (60-month) mark, delivered by a standard blended team of approximately 8-10 resources. Every currency figure is derived from the rate card and the effort plan below, so the totals reconcile across all tables.

Commercial validity period, price escalation, retention, performance guarantees and any exchange-rate assumptions must still be completed exactly as required by the tender conditions before the figures are transferred to the official Annexure L pricing forms. All estimates are benchmark-based and subject to T2 Technologies' final review.

17.1 Pricing Basis and Assumptions

- All rates are South African enterprise-IT top-of-band (upper-quartile) benchmark rates (2026), quoted per hour and per day, and must be confirmed against T2 Technologies' actual rate card before submission.
- Prices exclude VAT; VAT at 15% is shown separately in the five-year summary.
- Software is licensed as T2's proprietary SaaS platform on a named-user basis: 20 users at R 3 600 per user per month.
- The implementation is a fixed-price engagement invoiced against the delivery milestones in 17.7; effort is shown for transparency and change-control.
- Support & maintenance begins at production go-live (a three-month warranty is included in Stabilisation) and runs to the five-year mark.
- Third-party cloud/hosting, network connectivity, non-standard integrations, hardware and travel/ disbursements outside Gauteng are excluded and quoted at cost on confirmation.

17.2 Software / Subscription Licence

Item	Qty	Unit Basis	Annual (excl. VAT)	5-Year (excl. VAT)
T2 FBM SaaS — named-user licence	20	R 3 600 / user / month	R 864 000	R 4 320 000

17.3 Professional Services Rate Card

The following top-of-band (u
implementation and suppor

Role	Hourly (excl. VAT)	Day Rate (excl. VAT)
Programme / Project Manager	R 1 500	R 12 000
Solution Architect	R 1 850	R 14 800
Business Analyst	R 1 250	R 10 000
Senior Software Engineer	R 1 400	R 11 200
Software Engineer	R 1 000	R 8 000

Role	Hourly (excl. VAT)	Day Rate (excl. VAT)
QA / Test Analyst	R 950	R 7 600
Data Migration Specialist	R 1 300	R 10 400
DevOps / Cloud Engineer	R 1 400	R 11 200
Change & Training Specialist	R 1 050	R 8 400

17.4 Implementation Work-Package Pricing

Fixed-price implementation broken

Ref	Work Package	Effort (p-days)	Price (excl. VAT)
WP1	Business Process Analysis	170	R 1 884 000
WP2	Solution Architecture & Design	225	R 2 952 000
WP3	Build, Configure & Test	820	R 7 416 000
WP4	Data Migration	260	R 2 448 000
WP5	Deployment	155	R 1 646 000
WP6	Stabilisation & Go-Live (incl. 3-month warranty)	205	R 1 958 000
WP7	Training & Knowledge Transfer	140	R 1 216 000
WP8	Change Management	170	R 1 564 000
WP9	Project & Programme Management	320	R 3 896 000
	Total implementation (fixed price)	2465	R 24 980 000

17.5 Support and Maintenance

Item	Basis	Annual (excl. VAT)	Term	Total (excl. VAT)
Managed support & maintenance	Blended support team	R 3 330 400	4 yrs (post go-live)	R 13 321 600

Support covers L2/L3 application s management under an agreed SL

17.6 Five-Year Total Cost of Ownership

Year	Implementation	Licence	Support & Maint.	Annual Total
Year 1	R 24 980 000	R 864 000	—	R 25 844 000
Year 2	—	R 864 000	R 3 330 400	R 4 194 400
Year 3	—	R 864 000	R 3 330 400	R 4 194 400

Year	Implementation	Licence	Support & Maint.	Annual Total
Year 4	—	R 864 000	R 3 330 400	R 4 194 400
Year 5	—	R 864 000	R 3 330 400	R 4 194 400
Total (excl. VAT)				R 42 621 600
VAT @ 15%				R 6 393 240
Total (incl. VAT)				R 49 014 840

17.7 Payment Milestone Schedule

Implementation value is invoiced
month term.

Milestone	% of Implementation	Value (excl. VAT)
Contract signature & mobilisation	10%	R 2 498 000
Business Process Analysis sign-off	10%	R 2 498 000
Solution design approval	15%	R 3 747 000
Build, configure & test complete	25%	R 6 245 000
Data migration & UAT sign-off	15%	R 3 747 000
Production go-live	15%	R 3 747 000
Stabilisation & final acceptance	10%	R 2 498 000
Total	100%	R 24 980 000